



Slower ripening and enhanced antioxidant retention: Effects of crude bacterial IAA and temperature phasing on postharvest quality of 'Kluai Khai' (*Musa acuminata*, AA Group) bananas

 Hagi Sujateruna¹

 Savitr Trakulnaleamsai²

 Anchaya Mongkolchaiyaphruek^{3,*}

¹ Tropical Agriculture International Program, Faculty of Agriculture, Kasetsart University, Bangkok 10900, Thailand

² School of Integrated Science, Kasetsart University, Bangkok 10900, Thailand

³ Department of Horticulture, Faculty of Agriculture, Kasetsart University, Bangkok 10900, Thailand

 *Corresponding author: agrctt@ku.ac.th

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Abstract

Background and Objective: Postharvest deterioration after ripening severely limits the marketability of 'Kluai Khai' banana (*Musa acuminata*, AA group). This study evaluated whether an indole-3-acetic acid (IAA)-containing, cell-free solution produced by *Micrococcus yunnanensis* P2-23 and temperature phasing (TP) could maintain postharvest quality.

Methodology: A 4 × 2 factorial completely randomized design with three biological replicates combined with four nominal Salkowski-estimated IAA-equivalent concentrations (0, 250, 500, and 1,000 ppm) with non-TP and TP regimes. After a 3-min dip and storage at 12 ± 2°C for 14 days, fruit were ripened either continuously at 25°C or under TP (25, 22, 20, and 18°C on days 1–4, respectively, followed by 25°C). Visual quality, physicochemical characteristics, CO₂ production, and antioxidant-related attributes were evaluated through day 10.

Main Results: All fruits were fully yellow by day 6, with no treatment differences in final total soluble solids. By day 10, TP-treated fruit had brighter peel, higher titratable acidity and firmness (118.0–132.7 versus 56.9–66.6 N), and lower weight loss (5.55–7.03 versus 8.28–9.13%) than non-TP fruit. The 1,000 ppm crude solution

under TP maintained visual acceptability four days longer than the untreated non-TP control. TP also maintained higher late-stage ferric-reducing antioxidant power and total phenolic content, whereas non-TP fruit had higher DPPH activity. Respiration and crude-solution effects varied with time and concentration, with no consistent additional benefit or general synergy. HPLC-measured authentic IAA was 43.86% of the Salkowski estimate.

Conclusions: TP most consistently slowed late-stage deterioration without disrupting ripening and is therefore the best-supported practical treatment for ‘Kluai Khai’ bananas. The crude bacterial solution provided no consistent additional benefit, and its effects cannot be assigned solely to IAA because authentic IAA was lower than the Salkowski estimate and other metabolites may have been present.

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INTRODUCTION

Postharvest losses of fresh climacteric fruits remain a major challenge in the horticultural industry because rapid ripening, senescence, microbial decay, and physiological deterioration reduce storage life and marketability. This problem is particularly important for banana, in which the climacteric rise in respiration and ethylene production accelerates chlorophyll degradation, tissue softening, membrane deterioration, and reactive oxygen species (ROS) accumulation (Srivastava and Handa, 2005). ‘Kluai Khai’ (Sucrier; *Musa acuminata*, AA group) is an economically important diploid dessert banana cultivated commercially in Thailand. It is valued for its small, attractive golden-yellow fruit, thin peel, sweet taste, distinctive aroma, and suitability for fresh consumption and processing. It also contains phenolic compounds and flavonoids that contribute to its antioxidant properties

(Youryon and Supapvanich, 2016). Despite these desirable economic, sensory, and nutritional attributes, ‘Kluai Khai’ is highly perishable because of its rapid climacteric ripening, peel browning, softening, and development of senescence-associated spotting under ambient conditions (Da *et al.*, 2025). These characteristics limit its storage, distribution, and marketability (Ketsa, 2000). Therefore, postharvest technologies that delay ripening and senescence while preserving physicochemical and antioxidant-related quality are needed to reduce losses and extend the marketable life of this distinctive banana cultivar.

Conventional postharvest technologies, including low-temperature storage, modified-atmosphere packaging, and synthetic chemical treatments, are commonly used to delay ripening and extend the shelf life of horticultural products. However, increasing concerns regarding chemical residues, environmental

sustainability, and consumer safety have stimulated interest in physical and biologically based alternatives. Among physical postharvest technologies, temperature management is particularly important because temperature affects respiration, ethylene biosynthesis and responsiveness, membrane stability, pigment metabolism, cell-wall degradation, carbohydrate conversion, water loss, and oxidative reactions. Within a physiologically safe range, reducing the storage temperature suppresses metabolic activity and delays chlorophyll degradation, starch-to-sugar conversion, softening, water loss, and senescence. Nevertheless, exposure to temperatures below the chilling threshold can disrupt membrane integrity, induce oxidative stress and peel browning, interfere with normal pigment and carbohydrate metabolism, and cause uneven ripening after transfer to ambient conditions. Conversely, excessively high temperatures accelerate water loss and senescence and may impair normal peel de-greening. Effective temperature management must therefore balance metabolic suppression against the risks of chilling injury and heat-induced ripening disorders.

Temperature history can also exert persistent effects after fruit is transferred to a different temperature. Appropriate temperature management suppresses respiration and ethylene-associated ripening processes, thereby delaying peel browning, maintaining firmness, and extending banana shelf life (Seymour, 1993). Furthermore, storage of mature-green bananas at 13°C altered starch-granule structure and amylolytic metabolism during subsequent ripening at 19°C, demonstrating that temperature exposure can affect not only the rate but also the subsequent metabolic pattern of ripening (Peroni-Okita *et al.*, 2013). These observations provide a physiological basis for investigating temperature phasing (TP), defined in the present study as the sequential adjustment of ripening temperature through a short step-down regime

followed by transfer to ambient ripening conditions. Compared with continuous ripening at a single ambient temperature, TP may allow normal ripening to begin before temporarily reducing metabolic activity, thereby slowing respiration, fruit softening, water loss, and the progression of senescence. Such a regime may help coordinate the rate of ripening with the desired marketing period while maintaining normal peel-color development, texture, carbohydrate metabolism, and preserving antioxidant-related quality. However, the effectiveness of this approach is likely to depend on the cultivar and the specific temperature sequence employed, and its application to the highly perishable 'Kluai Khai' banana has not been adequately investigated.

A complementary biological approach may be provided by microbial products containing plant growth-regulating compounds. Plant growth-promoting bacteria produce a range of bioactive metabolites that can influence plant growth, stress responses, and metabolism. Among these metabolites, indole-3-acetic acid (IAA), the principal naturally occurring auxin, plays an important role in plant development and stress regulation (Spaepen and Vanderleyden, 2011). Microbial production of IAA may offer a renewable, biologically based source of auxin-related activity. In particular, the use of a cell-free crude bacterial solution could potentially reduce the purification, solvent, energy, and waste requirements associated with production of a purified compound. Nevertheless, such a solution may contain other bacterial metabolites in addition to authentic IAA, and its biological effects must therefore be interpreted as those of the crude solution as a whole.

The halotolerant bacterium *Micrococcus yunnanensis* P2-23 is a promising source of an IAA-containing bacterial solution because it can produce relatively high levels of IAA under saline and alkaline culture conditions (Chanram *et al.*, 2016). Its

tolerance to these conditions may support robust fermentation and the production of a metabolite-rich cell-free solution. Previous work demonstrated that a crude bacterial solution standardized according to its nominal IAA-equivalent concentration reduced flower drooping and color change and prolonged the vase life of *Dendrobium* flowers (Inoiew *et al.*, 2021). Although this finding provides preliminary evidence that a bacterial solution with auxin-related activity can influence postharvest senescence, its relevance to harvested edible climacteric fruits remains uncertain. The postharvest responses of banana may differ substantially because ripening is strongly regulated by autocatalytic ethylene production and involves coordinated changes in respiration, peel color, carbohydrate metabolism, firmness, and antioxidant systems.

Auxin provides a plausible physiological basis for modifying banana ripening. Auxin levels change during fruit development and ripening, and exogenous IAA can delay several ripening-associated processes. Application of IAA to pre-climacteric banana slices inhibited β -amylase activity, starch degradation, and sucrose formation, demonstrating that auxin can directly alter carbohydrate metabolism during banana ripening (Purgatto *et al.*, 2001). At the regulatory level, auxin–ethylene crosstalk may influence the transition from the pre-climacteric to the climacteric stage. Auxin signalling through TIR1/AFB receptors, Aux/IAA repressors, and auxin response factor transcription factors can modify ethylene biosynthesis and responsiveness, whereas the autocatalytic ethylene system coordinates peel de-greening, respiration, tissue softening, and senescence (Barry and Giovannoni, 2007; Li *et al.*, 2021; Tipu and Sherif, 2024). Auxin signalling is also interconnected with ROS metabolism, antioxidant defence systems, and phenylpropanoid biosynthesis, providing a possible link between auxin-related activity and the

maintenance of antioxidant compounds during ripening (Tipu and Sherif, 2024). However, the response to exogenous auxin depends on developmental stage, concentration, tissue type, and treatment timing. Auxin may delay the transition to ripening when applied before the climacteric rise, but it may not arrest ripening once autocatalytic ethylene signalling has become fully established.

Although exogenous IAA and low-temperature storage have each been investigated for their ability to modify banana ripening, several important knowledge gaps remain. Most studies of bacterial IAA have focused on plant growth, root development, and tolerance to abiotic stress, whereas its potential application to harvested climacteric fruit has received limited attention. In particular, the postharvest effects of an IAA-containing, cell-free crude solution produced by *M. yunnanensis* P2-23 have not been characterized adequately in ‘Kluai Khai’ bananas. In addition, previous temperature-management studies have predominantly employed constant storage or ripening temperatures, and little is known about the effectiveness of a sequential TP regime for this cultivar. It therefore remains unclear whether an IAA-containing crude bacterial solution and TP act independently, complementarily, or interactively in delaying ripening-associated deterioration and preserving physicochemical and antioxidant-related quality. Because the crude solution may contain metabolites other than authentic IAA, the present study evaluates the biological effects of the concentration-standardized crude solution and does not attempt to attribute the observed responses exclusively to IAA.

We hypothesized that TP and the IAA-containing crude bacterial solution would delay ripening-associated senescence and maintain the postharvest quality of ‘Kluai Khai’ bananas, and that their combined application might provide greater

quality retention than either treatment alone. Therefore, this study aimed to determine whether an IAA-containing, cell-free crude solution from *M. yunnanensis* P2-23 and a step-down TP regime could delay ripening and senescence and preserve the postharvest quality of 'Kluai Khai' bananas. Specifically, the main and interaction effects of crude bacterial solution concentration and TP on visual quality, physicochemical characteristics, respiratory behaviour, and antioxidant-related attributes were evaluated during postharvest ripening.

MATERIALS AND METHODS

Plant Materials

Green, physiologically mature, export-grade 'Kluai Khai' bananas (*Musa acuminata*, AA group) at ripening stage 1 (uniformly green peel with no visible yellowing) and approximately three-quarters full maturity stage, as indicated by fruit angularity, were sourced from a commercial grower in Chanthaburi province, Thailand. The banana bunches were harvested in the early morning and de-handled using a sanitized stainless-steel knife. The hands were then packed in commercial banana transport baskets and transported to the laboratory in a temperature-controlled vehicle maintained at 25°C. Upon arrival at the laboratory, banana hands showing visible mechanical damage or fungal infection were discarded. Only hands that were uniform in size, shape, and maturity were selected. All selected hands were washed with tap water, surface-sanitized with 250 ppm sodium hypochlorite, and dipped in a 500 ppm benzimidazole fungicide solution for 3 min to minimize fungal decay during the extended storage period. To facilitate uniform treatment application, replication, and destructive sampling while minimizing variation in fruit size, maturity, and exposure to bacterial IAA among experimental units, the air-dried banana hands were

divided into clusters of four fingers each. The clusters were subjected to their assigned treatments.

Crude Bacterial IAA Preparation

The halotolerant bacterium *Micrococcus yunnanensis* P2-23 was cultured in Haloalkaliphile medium (HA) at pH 8, supplemented with 30 mM L-tryptophan in a 1.5-L fermenter. The culture conditions were maintained at 35°C with an agitation rate of 260 rpm and an aeration rate of 1 L min⁻¹. The bacterial culture was incubated for 96–120 h. Bacterial growth was monitored by measuring the optical density at 600 nm (OD₆₀₀). The culture broth (pH 7.53) was then centrifuged to separate the cell pellet, and the cell-free supernatant containing IAA together with other extracellular bacterial metabolites was collected and designated the crude bacterial IAA solution. The crude bacterial IAA solution was sterilized by autoclaving at 121°C for 15 min before subsequent analyses. Autoclaving was employed based on previous reports demonstrating that IAA remained stable or detectable following conventional autoclaving and that IAA-containing cell-free supernatant and crude extracts from another *Micrococcus* species retained measurable IAA-equivalent content after treatment at 121°C for 15 min (Yamakawa *et al.*, 1979; Boonmahome and Mongkoltharuk, 2023). All crude bacterial IAA solutions used in the experiment were prepared from a single fermentation batch of *Micrococcus yunnanensis* P2-23 to ensure consistency among treatments. Following autoclaving, the sterile stock solution, with a pH of 7.45, was diluted with distilled water to obtain nominal Salkowski-estimated IAA-equivalent concentrations of 250, 500, and 1,000 ppm. The final pH values of the respective treatment solutions were 7.47, 7.48, and 7.45, respectively.

Quantification of IAA-Equivalent Concentration

The IAA-equivalent concentration of the cell-free bacterial supernatant, pH 7.45, was initially estimated using the Salkowski colorimetric assay (Gordon and Weber, 1951). Briefly, the appropriate volume of cell-free supernatant was reacted with Salkowski reagent, and the absorbance was measured at 530 nm. The IAA-equivalent concentration was calculated using a calibration curve prepared with IAA standard. Because the Salkowski assay is not specific exclusively to IAA, the resulting concentrations were expressed as Salkowski-estimated IAA-equivalent concentrations. Aliquots of the cell-free supernatant collected immediately before and after autoclaving at 121°C for 15 min were analysed to evaluate changes associated with heat sterilization. The pH of each aliquot was measured using a calibrated pH meter. The Salkowski assay was performed directly on the crude cell-free supernatant, whereas HPLC analysis was conducted using the corresponding ethyl acetate-extracted fractions.

HPLC Analysis of IAA

For HPLC analysis, IAA was extracted from the cell-free bacterial supernatant by liquid–liquid extraction using ethyl acetate. The ethyl acetate-extracted fraction was analyzed using a Shimadzu LC-10A liquid chromatograph equipped with an Inertsil ODS-3 column (4.6 × 250 mm, 5 µm particle size) maintained at 30°C. The mobile phase consisted of 0.8% acetic acid and acetonitrile at a ratio of 55:45 (v/v), delivered at a flow rate of 1 mL min⁻¹. IAA was detected using a photodiode array detector at 230 nm. IAA was identified by comparing the retention time of the sample peak with that of an IAA standard. The retention time of the IAA standard was 5.20–5.23 min, while the corresponding sample peak was detected at 5.22–5.23 min. Quantification was

performed using an IAA calibration curve over the concentration range of 31.25–1,000 µg mL⁻¹. The calibration curve was constructed by relating chromatographic peak area (x) to IAA concentration (y), according to the equation $y = 0.000355364x - 14.7623$, with a coefficient of determination (R^2) of 0.998376. Samples were appropriately diluted to fall within the calibration range before HPLC analysis, and the measured concentrations were corrected using the corresponding dilution factors to calculate the IAA concentration in the original cell-free supernatant.

IAA and Temperature Phasing Treatments

The experiment was conducted in a 4 × 2 factorial in a completely randomized design (CRD). The two factors were four nominal concentrations of the crude bacterial IAA solution (0, 250, 500, and 1,000 ppm, based on Salkowski-estimated IAA equivalents) and two temperature conditions (non-TP and TP), resulting in eight treatment combinations. Each treatment combination consisted of three independent biological replicates, with one basket representing one experimental unit. Each basket contained ten four-finger clusters. The experiment therefore comprised twenty-four baskets containing 240 clusters, with 12 baskets assigned to each temperature condition. The banana clusters were randomly dipped in their respective crude bacterial IAA solutions for 3 min and then air-dried at room temperature. Clusters assigned to the 0 ppm treatment were dipped in distilled water for the same duration. The treated clusters were allocated among polyethylene (PE)-bag-lined baskets, with ten clusters placed in each basket. Each basket was loosely covered with the PE bag without sealing. The baskets were then stored at 12 ± 2°C for 14 days to simulate the refrigerated transportation and distribution period

commonly experienced by mature-green export bananas before subsequent ripening and retail handling. At the end of the 14-day storage period, the baskets were removed from cold storage, and the banana clusters were visually inspected for chilling injury. No visible symptoms, including peel browning, surface pitting, or abnormal peel discoloration, were observed. Twelve baskets were assigned to the non-TP condition, and the remaining twelve to the TP condition. Under non-TP condition, the bananas were maintained continuously at 25°C for a 5-day ripening period followed by a 5-day storage period (10 experimental days). Under TP condition, the bananas were maintained at 25°C on day 1, 22°C on day 2, 20°C on day 3, and 18°C on day 4. On day 5, they were transferred to 25°C and maintained at this temperature for the remainder of the experiment. Measurements were performed on days 0, 1, 6, 8, and 10, with days 0 and 1 defined as the day after harvest and the day after pre-storage, respectively. At each sampling date, two clusters (eight banana fingers) were destructively sampled from each basket. Consequently, six clusters were sampled from each treatment combination at each sampling date. The two clusters collected from the same basket on a given sampling date were regarded as subsamples, and their measurements were averaged to obtain one value for each biological replicate before statistical analysis. Over the five sampling dates, all ten clusters in each basket were destructively sampled.

Determination of Physicochemical Quality

Fruit ripening was evaluated using a four-point scale based on the 'Kluai Khai' banana ripening chart: 1 = green (unripe), 2 = light green, 3 = yellow (ripe), and 4 = yellow with brown spots (overripe) (Tongpoolsomjit *et al.*, 2020). Peel color was measured at the midsection of each fruit using a colorimeter (MiniScan XE Plus, HunterLab, USA). Six banana

fingers from six clusters across the three biological replicates were randomly selected and monitored at all five evaluation times. The instrument recorded the International Commission on Illumination (CIE) color parameters L^* , a^* , and b^* , where L^* represents lightness, negative and positive a^* values indicate greenness and redness, respectively, and positive b^* values indicate yellowness.

At each of the five evaluation times, 12 banana fingers were randomly selected for destructive analyses from six clusters within each treatment combination. Fruit firmness was measured once at the midsection of each finger, with the peel intact, using a firmness tester (Effegi Bishop FT 327, Alfonsine, Italy) fitted with a 0.3-cm-diameter probe. The four measurements within each biological replicate were averaged, and firmness was expressed in newtons (N). Total soluble solids (TSS) content of the pulp was determined using a hand-held refractometer (Atago N1, Japan) and expressed as °Brix. Titratable acidity (TA) was determined by titration with 0.01 N NaOH using phenolphthalein as an indicator, and the results were expressed as a percentage of malic acid, the predominant organic acid in ripe banana pulp (Maduwanthi and Marapana, 2019).

Weight loss was calculated as the percentage reduction relative to the initial weight of six banana clusters. Sample weights were recorded using an analytical balance (Mettler NewClassic SG Balance, Switzerland). Respiration rates were measured in three banana clusters using a gas analyzer (Mocon Dansensor CheckPoint 3, Denmark) and calculated according to the method of Kays (1991). The rates of O_2 consumption (equation 1) and CO_2 evolution (equation 2) were determined as follows:

$$RO_2 = \frac{(P_i O_2 - P_f O_2) \times V_f}{100 \times W \times (\Delta t)} \quad \text{--- (1)}$$

$$RCO_2 = \frac{(P_i CO_2 - P_f CO_2) \times V_f}{100 \times W \times (\Delta t)} \quad \text{--- (2)}$$

In the equations, RO_2 and RCO_2 represent O_2 consumption and CO_2 production rates ($mL\ kg^{-1}\ h^{-1}$), respectively. P_i and P_f are the initial and final headspace gas concentrations (% v/v), V_f is the volume of the closed chamber (mL), W is the sample weight (kg), and $\Delta t = t_f - t_i$ is the closure duration (h).

Determination of Antioxidant Quality

To minimize sample variation, banana pulp from each of the three biological replicates was halved, pooled, and weighed to obtain 3 g of fresh tissue. Three independent extractions were then performed for each biological replicate. The pulp was ground using a mortar and pestle with the frequent addition of liquid nitrogen until a fine powder was obtained. The powdered sample was homogenized in 20 mL of 80% ethanol and centrifuged at $20,000 \times g$ for 20 min at $4^\circ C$ to obtain the supernatant containing the bioactive compounds. Free radical scavenging activity was determined using the 2,2-diphenyl-1-picrylhydrazyl (DPPH) assay (Saeed *et al.*, 2012). A stock solution was prepared by dissolving 12 mg of DPPH in 50 mL of 95% methanol. The working solution was prepared by diluting the stock solution with methanol to obtain an absorbance of approximately 0.98. Subsequently, 2.4 mL of the working solution was mixed with 0.1 mL of the sample supernatant and incubated in the dark at room temperature for 30 min. The absorbance of three replicates was then measured at 517 nm using a spectrophotometer (PG Instrument, T60 UV-Visible). DPPH radical scavenging activity (equation 3) was calculated using the following formula:

Scavenging activity (%) =

$$\frac{(\text{Abs}_{517\text{ control}} - \text{Abs}_{517\text{ sample}})}{\text{Abs}_{517\text{ sample}}} \times 100 \quad \text{--- (3)}$$

The antioxidant capacity of banana pulp was determined using the ferric reducing antioxidant power (FRAP) assay (Benzie and Strain, 1996). The

supernatant obtained from the extraction procedure described above was used for the analysis. The FRAP reagent was freshly prepared by mixing 300 mM acetate buffer (pH 3.6), 10 mM TPTZ (2,4,6-tri(2-pyridyl)-s-triazine), and 20 mM ferric chloride hexahydrate in a ratio of 10:1:1 (v/v/v). The reagent was pre-warmed at $37^\circ C$ for 4 min before use. For the assay, 2.85 mL of the FRAP reagent was mixed with 0.15 mL of the sample extract and incubated in the dark at room temperature for 30 min. Absorbance was measured at 593 nm using a spectrophotometer. A standard curve was prepared simultaneously using ferrous sulfate solutions at concentrations ranging from 0 to 2,000 μM and processed under the same conditions. Antioxidant capacity from three replicates was quantified and expressed as mg ferrous sulfate equivalents per 100 g fresh weight (mg FSE/100 g FW).

Total phenolic content (TPC) was determined using the Folin-Ciocalteu reagent (FCR) method (Singleton *et al.*, 1999) using the same supernatant extract described above. Briefly, 0.2 mL aliquot of 0.25 N Folin-Ciocalteu reagent was diluted with 2.4 mL distilled water and mixed with 0.1 mL of the sample extract. Subsequently, 0.3 mL of 1 N sodium carbonate was added, and the mixture was thoroughly stirred. The reaction mixture was incubated at $30^\circ C$ for 60 min, cooled on ice ($0^\circ C$) for 60 min, and then maintained in the dark at room temperature for 15 min. Absorbance was measured at 760 nm using a spectrophotometer. Total phenolic content from three replicates was represented as milligrams of gallic acid equivalent per 100 grams of fresh weight (mg GAE/100 g FW), based on a gallic acid standard curve ranging from 0 to 100 $mg\ L^{-1}$.

Statistical Analysis

Data from each sampling date were analyzed separately using two-way analysis of variance (ANOVA) based on the factorial model presented in equation 4.

Results are presented as the mean $\pm$ standard deviation. When significant treatment effects were detected, mean comparisons were performed using the least significant difference (LSD) test. Statistical analyses were conducted using SPSS software (version 25; IBM Corp., USA), and differences were considered statistically significant at $P < 0.05$.

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk} \quad \text{--- (4)}$$

where y_{ijk} is the observed value of the response variable for the k^{th} replicate at the i^{th} IAA concentration and the j^{th} TP condition, μ is the overall mean, A_i is the IAA concentration effect, B_j is the TP effect, AB_{ij} is the interaction effect between IAA concentration and TP effect, and ε_{ijk} is the random experimental error. The subscripts i , j , and k represented the levels of factor A, factor B, and replications, respectively.

RESULTS AND DISCUSSION

Visual Appearance and Peel Color

Visual observations showed that all fruit, regardless of treatment, reached ripening stage 2 (light-green peel) by day 1 and stage 3 (fully yellow peel) by day 6 (Figure 1). Neither the nominal IAA-equivalent concentration of the crude bacterial solution nor TP therefore interfered with normal peel-color development. Treatment differences became evident during the later stages of storage, mainly after the fruit had turned yellow, indicating an effect on late-stage visual quality rather than on the normal progression of peel ripening. Brown spotting was visible in the untreated non-TP control by day 6, whereas fruit treated with the 1,000 ppm crude bacterial solution under TP remained largely uniform yellow, with only slight browning, through day 10. Thus, the combined treatment delayed pronounced brown spotting and extended the visually acceptable period by approximately 4 days compared with the control. Specifically, 'Kluai Khai' banana fruit held at 25°C for 8 days typically exhibits skin blackening and

senescence spots (Youryon and Supapvanich, 2016). Brown spotting is a characteristic feature of banana-peel senescence and has been associated with phenolic oxidation by polyphenol oxidase (PPO), resulting in quinone formation and brown pigment accumulation (Yang *et al.*, 2001; Nguyen *et al.*, 2004). These observations are therefore consistent with delayed visible senescence and may be associated with reduced oxidative enzyme activity. From a practical perspective, all fruit reached the fully yellow stage, corresponding to eating ripeness, by day 6. However, the combined treatment maintained acceptable visual quality through days 8–10, suggesting that it may extend the visually marketable period beyond the onset of eating ripeness.

Changes in the peel color parameters (L^* , a^* , and b^*) of 'Kluai Khai' bananas over the 10-day storage period are presented in Table 1. At day 0 (after harvest), all fruit samples exhibited identical initial values across treatment groups: $L^* = 61.5 \pm 1.74$, $a^* = -9.3 \pm 0.53$, and $b^* = 31.5 \pm 1.10$, representing bright, green, and moderately yellow peel characteristics. On day 10, TP-treated bananas exhibited significantly higher L^* values than non-TP fruits across all crude bacterial IAA concentrations (Table 1). The highest L^* values were recorded in fruits treated with TP combined with 500 and 1,000 ppm crude IAA (68.83 and 68.22, respectively), with no significant difference observed between these two treatments. Conversely, the untreated non-TP control exhibited the lowest L^* value (47.15), which was significantly lower than those of all TP-treated groups. Nevertheless, the difference between TP and non-TP fruits at 250 ppm IAA remained non-significant. These results demonstrated that TP was the primary factor driving the retention of peel brightness, with the highest L^* values observed when TP was combined with 500 or 1,000 ppm crude bacterial IAA. Both crude bacterial IAA concentration and TP significantly affected L^* values ($P < 0.01$), with



Figure 1 Visual appearance of 'Kluai Khai' bananas treated with different concentrations of crude bacterial IAA under temperature phasing (TP) and non-TP conditions during 10 days of storage

a significant IAA $\times$ TP interaction ($P < 0.05$), indicating that the response depended on the treatment combination. Therefore, the maintenance of higher L^* values in TP-treated fruits may reflect reduced peel darkening and slower late-stage progression toward over-ripening and visible senescence.

Both crude bacterial IAA concentration and TP had significant main effects on the green-to-red (a^*) coordinate ($P < 0.01$), with no significant interaction observed ($P > 0.05$). All treatments exhibited a transition from negative a^* values at day 0 (-9.3) to positive values at day 10, reflecting greenness loss driven by chlorophyll degradation, a primary process underlying peel color change during banana ripening (Gooding *et al.*, 2001). Nevertheless, higher crude bacterial IAA concentrations (500 and 1,000 ppm) combined with TP attenuated the rise in a^* values (8.73 ± 1.10 and 8.65 ± 0.84 , respectively) compared to the untreated non-TP control (12.50 ± 3.95), demonstrating a delayed de-greening process. Regarding peel yellowness, the b^* value at day 10 was significantly influenced by TP ($P < 0.01$), whereas crude bacterial IAA concentration had no significant effect ($P > 0.05$). TP-treated fruits maintained higher b^* values (37.24 – 38.90) compared to non-TP fruits (25.54 – 31.87), suggesting that TP maintained vibrant yellow peel pigmentation while preventing the dull, brownish discoloration observed in non-TP controls.

The delayed discoloration observed in TP-treated bananas is consistent with previous reports showing that gradual temperature reduction slows respiration rate, ethylene action, and chlorophyll degradation in climacteric fruits (Srivastava and Handa, 2005). In addition, exogenous IAA has been reported to delay ripening-associated physiological processes, including starch degradation and senescence, in banana fruit (Purgatto *et al.*, 2001).

Thus, these findings demonstrate that temperature phasing plays a primary role in

maintaining banana peel quality during ripening and storage. When combined with crude bacterial IAA (500–1,000 ppm), TP effectively retains peel brightness, mitigates chlorophyll degradation, and delays senescence-associated darkening. This superior performance highlights a complementary effect among hormonal regulation, bacterial metabolites, and temperature management. Mechanistically, the reduced browning in treated bananas may be attributed to lower PPO activity and slower oxidative metabolism, which are likely supported by an enhanced antioxidant defense system.

Total Soluble Solids Concentration, Titratable Acidity, and Firmness

The physicochemical properties of 'Kluai Khai' bananas were significantly affected by temperature phasing and crude bacterial IAA treatment during ripening (Table 2). Changes in total soluble solids, titratable acidity, and peel firmness reflected the progression of ripening and senescence during storage.

Total soluble solids increased markedly in all treatments from 4.20°Brix on day 0 to 17.20 – 19.40°Brix on day 10, indicating the normal conversion of starch into soluble sugars during banana ripening. However, no significant differences in TSS were observed among treatments at the end of storage, suggesting that neither crude bacterial IAA nor TP strongly altered sugar accumulation. The increase in TSS during ripening is a characteristic metabolic process in banana fruit and is associated with starch hydrolysis mediated by amylolytic enzymes. Purgatto *et al.* (2001) demonstrated that IAA treatment of unripe banana slices delayed the increase in β -amylase activity, starch degradation, and sucrose formation, providing evidence that auxin can influence carbohydrate metabolism during ripening. Nevertheless, in the present study, the TSS values remained unaffected, indicating that while crude bacterial IAA and TP

delayed external senescence symptoms, the fruits still underwent normal ripening development.

Titrateable acidity increased during storage in all treatments, ranging from 0.29 to 0.39% malic acid on day 10. Significant main effects of crude bacterial IAA concentration ($P < 0.01$) and TP treatment ($P < 0.05$) were observed, with no significant IAA $\times$ TP interaction. Bananas treated with TP generally maintained higher acidity than non-TP fruits, particularly at 0, 250, and 1,000 ppm crude bacterial IAA, whereas identical values were observed at 500 ppm. The highest TA values (0.39%) were observed in TP-treated bananas at 0 and 250 ppm crude IAA, while the lowest value (0.29%) occurred in bananas treated with 1,000 ppm crude IAA without TP. The comparatively low TA observed at 1,000 ppm under both temperature conditions suggests that the significant concentration effect may have been associated with the response at the highest crude bacterial IAA concentration.

The increase in TA from the initial value to those observed on day 10 reflects a shift in organic acid balance as ripening progressed. Organic acid accumulation in fleshy fruit is governed by the balance among synthesis, degradation, and vacuolar storage, with phosphoenolpyruvate carboxylase (PEPC)-mediated carboxylation contributing to malate synthesis; organic acid metabolism also changes substantially during postharvest banana ripening (Etienne *et al.*, 2013). Auxin may interact with ethylene biosynthesis during climacteric ripening, and auxin has been reported to up-regulate climacteric ethylene synthesis in peach (Trainotti *et al.*, 2007). Thus, the comparatively low TA at 1,000 ppm crude bacterial IAA could be consistent with altered ripening-associated organic acid metabolism and possibly greater respiratory utilization of organic acids. Temperature can also influence fruit acidity by altering organic acid metabolism and vacuolar storage, and higher

temperatures have generally been associated with lower TA and reduced malate and citrate concentrations, including in banana (Etienne *et al.*, 2013). The higher overall TA observed under TP may therefore reflect the influence of the earlier lower-temperature phases on the balance between organic acid accumulation and utilization. Nevertheless, because TP-treated fruit were subsequently returned to 25°C from day 5 onward and the day-10 respiration data did not show a consistently lower respiration rate under TP, the higher TA cannot be attributed specifically to reduced respiratory consumption of organic acids. Overall, TP may have influenced organic acid balance during late-stage ripening.

Peel firmness showed the clearest response to TP treatment during storage ($P < 0.01$). All fruits exhibited substantial softening during ripening, with firmness declining from 348.0 N on day 0 (unripe stage) to 56.9–132.7 N on day 10, indicating substantial softening as ripening progressed. Nevertheless, TP-treated bananas consistently maintained significantly ($P < 0.01$) higher firmness (118.0–132.7 N) than non-TP fruits (56.9–66.6 N), regardless of crude bacterial IAA concentration. When averaged across IAA concentrations, firmness was approximately 123.1 N under TP and 59.5 N under non-TP conditions, indicating that TP-treated fruit remained more than twice as firm as fruit maintained continuously at 25°C.

In contrast, neither crude bacterial IAA concentration nor the IAA $\times$ TP interaction significantly affected firmness. Although the 1,000 ppm IAA + TP treatment exhibited the highest firmness (132.7 N), this treatment cannot be considered statistically superior to TP alone or to the other TP treatments because neither the crude bacterial IAA main effect nor the interaction was significant. Thus, the data identify TP, rather than crude bacterial IAA application, as the principal factor associated with firmness retention and provide no statistical evidence of a

complementary or synergistic effect of IAA and TP on this attribute.

This response is also consistent with other quality changes observed during storage. Although all fruit reached the fully yellow stage by day 6, indicating that TP did not inhibit the normal ripening progression, TP-treated fruit subsequently retained substantially greater firmness and exhibited lower weight loss (5.55–7.03%) than non-TP fruit (8.28–9.13%). Together, these findings suggest that the main benefit of TP was particularly effective in slowing texture deterioration during the later stages of ripening. The lower weight loss under TP may also have contributed to firmness retention through better maintenance of tissue water status.

Fruit softening during ripening is associated with coordinated changes in cell-wall structure and the modification of pectic and hemicellulose polysaccharides involving several cell-wall-modifying enzymes, including polygalacturonase, pectin methylesterase, and other hydrolases (Brummell and Harpster, 2001). The lower-temperature phases may therefore have slowed metabolic and cell-wall-modifying processes associated with softening, thereby contributing to the greater firmness retained after the fruit was returned to 25°C. However, because cell-wall composition and the activities or expression of softening-related enzymes were not measured, a direct effect of TP on these processes cannot be established from the present data.

Importantly, the maintenance of firmness and titratable acidity, together with delayed peel discoloration, demonstrates that TP was the principal treatment associated with maintaining postharvest quality and delaying visible senescence in 'Kluai Khai' bananas. Although crude bacterial IAA exhibited specific responses, no significant IAA × TP interaction was detected for firmness or titratable acidity. Thus, for these attributes, the response to TP did not depend

significantly on crude bacterial IAA concentration. The maintenance of firmness and visual quality is nevertheless commercially relevant because texture and appearance are important determinants of fruit acceptability and marketability during postharvest handling and distribution.

Weight Loss and CO₂ Production Rate

Weight loss of 'Kluai Khai' bananas increased progressively during ripening in all treatments (Table 3). Significant main effects of both crude bacterial IAA concentration and temperature phasing were observed, whereas the IAA × TP interaction was not significant. The effect of crude bacterial IAA concentration was significant on day 6 ($P < 0.05$) and highly significant on days 8 and 10 ($P < 0.01$), while TP had a highly significant effect on days 6, 8, and 10 ($P < 0.01$).

On day 10, TP-treated bananas exhibited markedly lower weight loss (5.55–7.03%) than non-TP fruit (8.28–9.13%). When averaged across crude bacterial IAA concentrations, weight loss was approximately 6.41% under TP and 8.75% under non-TP conditions, representing a reduction of approximately 26.7% under TP. The lowest weight loss was observed in TP-treated bananas without crude bacterial IAA application (5.55%), whereas the highest value occurred in non-TP bananas treated with a crude bacterial IAA concentration of 1,000 ppm (9.13%). These results identify TP as the principal treatment associated with reduced cumulative weight loss during storage.

The significant main effect of crude bacterial IAA concentration was associated with increased weight loss. When averaged across temperature conditions, day 10 weight loss increased from 6.92% at 0 ppm to 7.45, 7.98, and 7.97% at 250, 500, and 1,000 ppm, respectively. Thus, the overall response indicates greater weight loss at the higher concentrations of crude bacterial IAA. The IAA × TP

Table 1 Changes in peel color (L^* , a^* , b^*) of 'Kluai Khai' bananas treated with different concentrations of crude bacterial IAA under temperature phasing (TP) and non-TP conditions during 10 days of storage

Treatments		Lightness (L^*)		Green-to-red value (a^*)		Blue-to-yellow value (b^*)	
IAA conc. (ppm)	Temp. phasing	Day 0	Day 10	Day 0	Day 10	Day 0	Day 10
0	Non-TP	61.5 ± 1.74	47.15 ± 10.0 ^e	-9.3 ± 0.53	12.50 ± 3.95	31.5 ± 1.10	25.54 ± 9.46
	TP	61.5 ± 1.74	60.32 ± 2.8 ^{bc}	-9.3 ± 0.53	12.30 ± 1.61	31.5 ± 1.10	37.24 ± 2.72
250	Non-TP	61.5 ± 1.74	57.81 ± 5.6 ^{cd}	-9.3 ± 0.53	11.05 ± 0.81	31.5 ± 1.10	31.87 ± 3.87
	TP	61.5 ± 1.74	63.34 ± 3.0 ^{abc}	-9.3 ± 0.53	10.91 ± 0.98	31.5 ± 1.10	38.90 ± 2.21
500	Non-TP	61.5 ± 1.74	52.07 ± 14.9 ^{de}	-9.3 ± 0.53	10.64 ± 1.26	31.5 ± 1.10	27.10 ± 10.8
	TP	61.5 ± 1.74	68.83 ± 1.8 ^a	-9.3 ± 0.53	8.73 ± 1.10	31.5 ± 1.10	38.17 ± 2.32
1,000	Non-TP	61.5 ± 1.74	58.39 ± 3.9 ^{cd}	-9.3 ± 0.53	10.48 ± 1.16	31.5 ± 1.10	31.05 ± 1.81
	TP	61.5 ± 1.74	68.22 ± 1.4 ^{ab}	-9.3 ± 0.53	8.65 ± 0.84	31.5 ± 1.10	37.56 ± 0.59
Crude IAA		-	**	-	**	-	ns
TP		-	**	-	**	-	**
IAA × TP		-	*	-	ns	-	ns
CV (%)		-	11.81	-	15.86	-	16.46

Note: Means within the same column followed by different superscript letters (a–e) are significantly different according to the least significant difference (LSD) test at $P < 0.05$. Values are presented as mean ± standard deviation ($n = 3$). ** $P < 0.01$, * $P < 0.05$, ns = not significant. IAA conc. = IAA concentration, Temp. phasing = temperature phasing, CV = coefficient of variation.

Table 2 Physicochemical properties of 'Kluai Khai' banana treated with different concentrations of crude bacterial IAA under temperature phasing (TP) and non-TP conditions during 10 days of storage

Treatments		Total soluble solids (°Brix)		Titratable acidity (% malic acid)		Peel firmness (N)	
IAA conc. (ppm)	Temp. phasing	Day 0	Day 10	Day 0	Day 10	Day 0	Day 10
0	Non-TP	4.2 ± 0.0	19.0 ± 0.0	0.13 ± 0.00	0.35 ± 0.03	348.0 ± 21.0	56.9 ± 18.45
	TP	4.2 ± 0.0	18.6 ± 0.0	0.13 ± 0.00	0.39 ± 0.03	348.0 ± 21.0	118.7 ± 17.14
250	Non-TP	4.2 ± 0.0	19.4 ± 0.0	0.13 ± 0.00	0.35 ± 0.03	348.0 ± 21.0	57.6 ± 17.94
	TP	4.2 ± 0.0	19.2 ± 0.0	0.13 ± 0.00	0.39 ± 0.06	348.0 ± 21.0	118.0 ± 14.63
500	Non-TP	4.2 ± 0.0	19.0 ± 0.0	0.13 ± 0.00	0.36 ± 0.04	348.0 ± 21.0	56.9 ± 17.25
	TP	4.2 ± 0.0	18.6 ± 0.0	0.13 ± 0.00	0.36 ± 0.04	348.0 ± 21.0	122.9 ± 22.44
1,000	Non-TP	4.2 ± 0.0	19.2 ± 0.0	0.13 ± 0.00	0.29 ± 0.04	348.0 ± 21.0	66.6 ± 20.48
	TP	4.2 ± 0.0	17.2 ± 0.0	0.13 ± 0.00	0.32 ± 0.03	348.0 ± 21.0	132.7 ± 16.83
Crude IAA		-	-	-	**	-	ns
TP		-	-	-	*	-	**
IAA × TP		-	-	-	ns	-	ns
CV (%)		-	-	-	10.35	-	20.08

Note: Values are presented as mean ± standard deviation ($n = 3$). ** $P < 0.01$, * $P < 0.05$, ns = not significant. IAA conc. = IAA concentration, Temp. phasing = temperature phasing, CV = coefficient of variation.

interaction was not significant on any sampling date ($P > 0.05$), indicating that the reduction in weight loss associated with TP did not vary significantly among crude bacterial IAA concentrations. The TP retained its beneficial effect across all crude bacterial IAA concentrations, whereas the significant crude bacterial IAA main effect was associated with an overall increase in weight loss.

Postharvest weight loss primarily reflects water loss through transpiration, together with carbon loss resulting from respiratory substrate consumption. Excessive weight loss may contribute to peel shriveling, reduced tissue turgor, softening, and loss of marketability (Wills and Golding, 2016). Banana respiration is strongly influenced by storage temperature and atmospheric composition (Bhande *et al.*, 2008). Opara *et al.* (2013) similarly reported lower weight loss and slower firmness loss in 'Malindi' Cavendish bananas maintained under refrigerated storage conditions than in fruit stored under warmer conditions. However, their temperature, relative-humidity, and storage regimes differed from the TP treatment used in the present study.

The lower cumulative weight loss under TP may therefore reflect reduced transpiration and metabolic mass loss during the lower-temperature phases. However, the respiration measurements did not demonstrate a consistently lower respiratory rate under TP after the fruit were returned to 25°C. Consequently, the lower weight loss may be attributed to the cumulative outcome of temperature management across the entire storage period. The lower weight loss observed in TP-treated bananas was consistent with their higher peel-inclusive fruit firmness and better peel-color retention during late storage. These concurrent responses suggest that TP slowed several aspects of postharvest quality deterioration, although they do not establish a common physiological mechanism. Reduced weight loss may have contributed

to firmness retention through better maintenance of tissue turgor, but tissue water status was not measured directly. From a commercial perspective, limiting weight loss is important because excessive mass and moisture loss can reduce saleable weight, visual quality, and consumer acceptance (Wills and Golding, 2016). Accordingly, the present results support TP, rather than the combined application of TP and crude bacterial IAA, as the primary treatment for reducing weight loss and maintaining the postharvest quality of 'Kluai Khai' bananas.

Across days 1–10, CO₂ production of 'Kluai Khai' bananas generally increased as ripening progressed, although the effects of crude bacterial IAA and TP varied with storage duration (Table 4). This overall increase is consistent with the climacteric behaviour of banana fruit, in which respiration rises during ripening in association with ethylene-regulated biochemical changes (Golding *et al.*, 1999). On day 1, fruit assigned to the TP treatment had an approximately 12.7% lower mean CO₂ production rate than non-TP fruit. Crude bacterial IAA concentration also significantly affected respiration on this date; however, the absence of a significant interaction indicates that the low CO₂ production observed in the 1,000 ppm crude bacterial IAA + TP treatment cannot be attributed specifically to a combined effect of the two factors.

The respiratory response became concentration-dependent on day 8, as indicated by the significant crude bacterial IAA × TP interaction. TP reduced CO₂ production in fruit receiving 0 ppm crude bacterial IAA, whereas the responses at the other concentrations were variable and did not demonstrate consistent respiratory suppression by the combined treatments. By day 10, the pattern had shifted: TP-treated fruit exhibited an approximately 14.3% higher mean CO₂ production rate than non-TP fruit, regardless

of crude bacterial IAA concentration. Temperature is an important regulator of respiratory metabolism in banana fruit (Bhande *et al.*, 2008); nevertheless, the present TP treatment involved only temporary exposure to lower temperatures followed by a return to 25°C. Therefore, TP appeared to modify the temporal pattern of respiration rather than continuously suppress respiratory activity throughout storage.

Auxin-related signalling components also respond to ethylene during banana ripening, indicating potential crosstalk between the two hormonal pathways (Kong *et al.*, 2022). In the present study, however, the effect of crude bacterial IAA on respiration was transient and varied with concentration and storage stage. Because the bacterial preparation contained crude metabolites rather than purified IAA, these responses cannot be attributed exclusively to auxin. Overall, the results provide no evidence of a consistent combined effect of crude bacterial IAA and TP on respiration. The higher CO₂ production under TP on

day 10 may reflect a later respiratory rise, persistent metabolic activity following the return to 25°C, or differences in physiological stage; however, the available measurements are insufficient to confirm a delayed climacteric peak or delayed senescence.

The respiratory results should also be considered together with the other quality attributes. Despite their higher CO₂ production on day 10, TP-treated fruit retained greater peel-inclusive firmness, exhibited lower weight loss, and maintained better peel color than non-TP fruit. These concurrent responses are consistent with an alteration of late-stage ripening or senescence under TP, but they do not demonstrate that the higher respiration resulted from delayed senescence. More frequent measurements of both CO₂ and ethylene production, together with experiments using purified IAA and an appropriate bacterial-metabolite control, would be required to clarify the climacteric timing and the mechanism underlying the observed responses.

Table 3 Weight loss (%) of ‘Kluai Khai’ banana treated with different concentrations of crude bacterial IAA under temperature phasing (TP) and non-TP conditions during 10 days of storage

Treatments		Weight loss (%)			
IAA conc. (ppm)	Temp. phasing	Day 1	Day 6	Day 8	Day 10
0	Non-TP	0.00 ± 0.00	4.11 ± 0.56	6.23 ± 0.74	8.28 ± 0.92
	TP	0.00 ± 0.00	2.35 ± 0.63	3.63 ± 0.82	5.55 ± 0.89
250	Non-TP	0.00 ± 0.00	4.56 ± 0.50	6.41 ± 0.47	8.65 ± 0.50
	TP	0.00 ± 0.00	2.85 ± 0.30	4.48 ± 0.41	6.25 ± 0.52
500	Non-TP	0.00 ± 0.00	4.43 ± 0.52	6.69 ± 0.83	8.92 ± 0.91
	TP	0.00 ± 0.00	3.11 ± 0.27	5.17 ± 0.39	7.03 ± 0.50
1,000	Non-TP	0.00 ± 0.00	4.72 ± 0.34	6.98 ± 0.67	9.13 ± 0.89
	TP	0.00 ± 0.00	2.91 ± 0.54	4.87 ± 0.40	6.80 ± 0.42
Crude IAA		-	*	**	**
TP		-	**	**	**
IAA × TP		-	ns	ns	ns
CV (%)		-	0.56	0.70	0.78

Note: Values are presented as mean ± standard deviation (n = 3). **P < 0.01, *P < 0.05, ns = not significant.

IAA conc. = IAA concentration, Temp. phasing = temperature phasing, CV = coefficient of variation.

Table 4 CO₂ production rate (mL kg⁻¹ h⁻¹) of 'Kluai Khai' banana treated with different concentrations of crude bacterial IAA under temperature phasing (TP) and non-TP conditions during 10 days of storage

Treatments		CO ₂ production rate (mL kg ⁻¹ h ⁻¹)				
IAA conc. (ppm)	Temp. phasing	Day 0	Day 1	Day 6	Day 8	Day 10
0	Non-TP	34.13 ± 1.88	66.46 ± 15.2	117.31 ± 5.8	123.69 ± 6.4 ^a	113.99 ± 6.0
	TP	34.13 ± 1.88	59.09 ± 7.3	98.97 ± 2.1	102.10 ± 4.8 ^b	125.93 ± 1.3
250	Non-TP	34.13 ± 1.88	63.38 ± 8.2	104.08 ± 6.0	121.29 ± 7.5 ^a	115.27 ± 7.9
	TP	34.13 ± 1.88	58.52 ± 3.4	108.73 ± 8.1	108.91 ± 4.3 ^{ab}	135.05 ± 6.4
500	Non-TP	34.13 ± 1.88	79.51 ± 5.8	104.64 ± 9.1	121.71 ± 9.0 ^a	123.44 ± 16.4
	TP	34.13 ± 1.88	69.48 ± 8.1	101.25 ± 7.2	120.31 ± 7.3 ^a	136.96 ± 6.5
1,000	Non-TP	34.13 ± 1.88	64.81 ± 17.2	93.65 ± 22.0	104.44 ± 13.4 ^b	106.84 ± 2.1
	TP	34.13 ± 1.88	52.31 ± 7.5	104.51 ± 10.9	114.36 ± 21.5 ^{ab}	127.49 ± 20.4
Crude IAA		-	*	ns	ns	ns
TP		-	*	ns	ns	**
IAA × TP		-	ns	ns	*	ns
CV (%)		-	12.48	10.57	7.88	8.67

Note: Means within the same column followed by different superscript letters (a, b) are significantly different according to the least significant difference (LSD) test at $P < 0.05$. Values are presented as mean ± standard deviation ($n = 3$). ** $P < 0.01$, * $P < 0.05$, ns = not significant. IAA conc. = IAA concentration, Temp. phasing = temperature phasing, CV = coefficient of variation.

DPPH Scavenging Activity, Antioxidant Capacity, and Total Phenolic Content

Antioxidant-related responses of 'Kluai Khai' bananas varied according to the crude bacterial IAA concentration, temperature treatment, and storage duration (Tables 5–7). For DPPH radical-scavenging activity and ferric-reducing antioxidant power (FRAP), significant main effects of crude bacterial IAA concentration and TP, together with significant IAA × TP interactions, were detected on every sampling date from day 1 onward ($P < 0.01$). For total phenolic content (TPC), both main effects were significant throughout storage ($P < 0.01$), whereas the interaction was significant on days 1, 6, and 8 but not on day 10. These interactions indicate that the responses to TP depended on crude bacterial IAA concentration and storage stage; therefore, the results do not support a general conclusion that either treatment consistently enhanced all antioxidant-related measurements.

DPPH radical-scavenging activity increased from the initial value of 26.36% but did not increase progressively in every treatment (Table 5). On day 1, the highest value was observed in fruit treated with 1,000 ppm crude bacterial IAA under TP (50.03%), whereas subsequent responses differed among concentrations and temperature conditions. By day 8, non-TP fruit had higher DPPH activity than their corresponding TP-treated fruit at all crude bacterial IAA concentrations. The highest value on this date occurred in untreated non-TP fruit (58.20%), followed by the 250 ppm non-TP treatment (56.19%). On day 10, the highest value occurred in the 250 ppm non-TP treatment (54.35%), while TP-treated fruit remained lower than their corresponding non-TP groups. Thus, the DPPH results indicate a treatment- and stage-dependent response rather than a uniform increase during ripening or consistent enhancement by crude bacterial IAA or TP.

The higher DPPH activity in non-TP fruit during late storage could reflect changes in the abundance of particular radical-scavenging compounds as ripening advanced. Oxidative and antioxidative systems are known to change dynamically during climacteric-fruit ripening (Jiménez *et al.*, 2002), and ripening-stage-dependent changes in antioxidant measurements have also been reported in *Musa* AA bananas (Youryon and Supapvanich, 2017). However, the present DPPH data alone cannot demonstrate that non-TP fruit experienced greater oxidative stress, because ROS accumulation, lipid peroxidation, and antioxidant enzyme activities were not measured.

FRAP antioxidant capacity followed a different temporal pattern (Table 6). All treatments declined from the initial value of 158.77 mg FSE/100 g FW on day 0 to 86.17–122.30 mg FSE/100 g FW on day 1. The effect of TP remained concentration-dependent on day 6, but a clearer TP-associated response was evident during late storage. On day 8, TP-treated fruit had higher FRAP values than the corresponding non-TP fruit at every crude bacterial IAA concentration, with the highest value occurring in the 1,000 ppm + TP treatment (174.46 mg FSE/100 g FW). The same general pattern was observed on day 10, when TP-treated fruit ranged from 140.18 to 177.44 mg FSE/100 g FW, compared with 120.54–143.63 mg FSE/100 g FW in non-TP fruit. The highest day-10 value was recorded in untreated TP fruit (177.44 mg FSE/100 g FW).

These results show that TP was associated with higher ferric-reducing capacity during the late storage period, but they do not demonstrate a consistent additional benefit of crude bacterial IAA. In particular, untreated TP fruit had the highest FRAP value on day 10, and the significant IAA × TP interactions indicate that the effects of concentration differed among temperature conditions and sampling

dates. Consequently, the interaction should not be described as evidence that the combined treatment consistently preserved antioxidant capacity more effectively than either factor alone.

TPC initially declined from 28.66 mg GAE/100 g FW on day 0 in all treatments but generally increased during subsequent storage (Table 7). On day 8, TP-treated fruit contained 30.74–35.76 mg GAE/100 g FW, compared with 25.74–29.17 mg GAE/100 g FW in non-TP fruit. The significant interaction on this date indicates that the magnitude of the TP response depended on crude bacterial IAA concentration. On day 10, both crude bacterial IAA concentration and TP had significant main effects ($P < 0.01$), but their interaction was not significant. Averaged across concentrations, TP-treated fruit had a higher TPC than non-TP fruit (36.92 versus 31.83 mg GAE/100 g FW). The highest individual value was observed in the 1,000 ppm + TP treatment (38.77 mg GAE/100 g FW), followed by untreated TP fruit (38.36 mg GAE/100 g FW) and the 500 ppm + TP treatment (38.04 mg GAE/100 g FW). However, because the interaction was not significant, these differences do not provide evidence that higher crude bacterial IAA concentrations enhanced the TP response or produced a synergistic effect. The absence of an interaction indicates only that the effect of TP did not vary significantly among concentrations on day 10; it does not establish physiologically independent mechanisms.

Phenolic compounds contribute to the antioxidant properties of banana tissues (Someya *et al.*, 2002), but the higher TPC under TP cannot be attributed specifically to reduced phenolic oxidation. The observed values may reflect differences in phenolic synthesis and retention. Measurements of PPO activity and phenolic oxidation products would be required to distinguish phenolic accumulation from reduced oxidative degradation.

Table 5 DPPH scavenging activity (%) of 'Kluai Khai' banana treated with different concentrations of crude bacterial IAA under temperature phasing (TP) and non-TP conditions during 10 days of storage

Treatments		DPPH scavenging activity (%)				
IAA conc. (ppm)	Temp. phasing	Day 0	Day 1	Day 6	Day 8	Day 10
0	Non-TP	26.36 ± 2.1	38.61 ± 1.02 ^f	45.07 ± 0.12 ^d	58.20 ± 0.39 ^a	49.35 ± 0.52 ^b
	TP	26.36 ± 2.1	42.41 ± 0.29 ^e	42.99 ± 0.62 ^e	50.78 ± 0.66 ^c	32.41 ± 0.39 ^h
250	Non-TP	26.36 ± 2.1	37.93 ± 0.31 ^f	51.36 ± 0.39 ^a	56.19 ± 0.16 ^b	54.35 ± 0.97 ^a
	TP	26.36 ± 2.1	45.75 ± 0.65 ^{bc}	47.04 ± 0.20 ^c	51.43 ± 0.20 ^c	36.67 ± 0.72 ^f
500	Non-TP	26.36 ± 2.1	44.63 ± 0.51 ^d	49.73 ± 1.79 ^b	50.61 ± 0.74 ^c	46.21 ± 0.56 ^c
	TP	26.36 ± 2.1	46.77 ± 0.62 ^b	52.14 ± 0.54 ^a	41.16 ± 1.12 ^e	39.35 ± 0.56 ^e
1,000	Non-TP	26.36 ± 2.1	44.97 ± 0.48 ^{cd}	46.33 ± 0.18 ^{cd}	47.52 ± 0.51 ^d	43.37 ± 0.57 ^d
	TP	26.36 ± 2.1	50.03 ± 0.41 ^a	42.72 ± 0.39 ^e	40.82 ± 0.20 ^e	35.10 ± 0.20 ^g
Crude IAA		-	**	**	**	**
TP		-	**	**	**	**
IAA × TP		-	**	**	**	**
CV (%)		-	1.36	1.55	1.49	1.49

Note: Means within the same column followed by different superscript letters (a–h) are significantly different according to the least significant difference (LSD) test at $P < 0.05$. Values are presented as mean ± standard deviation ($n = 3$). ** $P < 0.01$. IAA conc. = IAA concentration, Temp. phasing = temperature phasing, CV = coefficient of variation.

Table 6 Antioxidant capacity (mg FSE/100 g FW) of 'Kluai Khai' banana treated with different concentrations of crude bacterial IAA under temperature phasing (TP) and non-TP conditions during 10 days of storage

Treatments		Antioxidant capacity (mg FSE/100 g FW)				
IAA conc. (ppm)	Temp. phasing	Day 0	Day 1	Day 6	Day 8	Day 10
0	Non-TP	158.77 ± 2.8	116.67 ± 2.4 ^b	117.75 ± 3.4 ^d	111.25 ± 2.5 ^e	133.60 ± 2.1 ^e
	TP	158.77 ± 2.8	104.56 ± 1.8 ^c	145.65 ± 3.8 ^a	131.92 ± 5.5 ^d	177.44 ± 4.3 ^a
250	Non-TP	158.77 ± 2.8	122.30 ± 2.5 ^a	112.67 ± 4.8 ^e	109.90 ± 3.7 ^e	120.54 ± 0.7 ^g
	TP	158.77 ± 2.8	86.17 ± 2.0 ^d	134.22 ± 3.3 ^b	143.39 ± 2.7 ^c	161.93 ± 3.5 ^b
500	Non-TP	158.77 ± 2.8	118.24 ± 2.3 ^b	122.44 ± 0.1 ^c	128.71 ± 3.9 ^d	124.94 ± 1.1 ^f
	TP	158.77 ± 2.8	117.66 ± 0.1 ^b	113.55 ± 1.6 ^e	143.98 ± 6.9 ^c	140.18 ± 0.8 ^d
1,000	Non-TP	158.77 ± 2.8	102.68 ± 0.3 ^c	104.46 ± 1.2 ^f	152.83 ± 2.7 ^b	143.63 ± 0.3 ^d
	TP	158.77 ± 2.8	86.51 ± 2.0 ^d	115.25 ± 1.8 ^{de}	174.46 ± 3.1 ^a	154.31 ± 4.4 ^c
Crude IAA		-	**	**	**	**
TP		-	**	**	**	**
IAA × TP		-	**	**	**	**
CV (%)		-	1.68	1.69	2.66	1.45

Note: Means within the same column followed by different superscript letters (a–g) are significantly different according to the least significant difference (LSD) test at $P < 0.05$. Values are presented as mean ± standard deviation ($n = 3$). ** $P < 0.01$. IAA conc. = IAA concentration, Temp. phasing = temperature phasing, CV = coefficient of variation.

The contrasting DPPH and FRAP patterns are not necessarily contradictory because these assays measure different chemical properties and vary in their sensitivity to individual antioxidant compounds. DPPH measures the ability of extractable compounds to neutralize a stable organic radical, whereas FRAP measures ferric-ion-reducing capacity. Consequently, the two assays do not necessarily respond in parallel (Prior *et al.*, 2005). Similar assay-dependent responses have been reported during the ripening of *Musa* AA bananas, in which total antioxidant capacity and TPC changed with ripening stage while DPPH activity exhibited a different pattern (Youryon and Supapvanich, 2017). In the present study, the pattern of lower DPPH activity but higher FRAP and TPC under TP during late storage suggests that TP altered the composition or relative contribution of extractable antioxidants rather than uniformly increasing or preserving all forms of antioxidant activity.

Exogenous IAA can influence banana-ripening metabolism. Purified IAA applied to pre-climacteric banana slices reduced the climacteric respiratory burst and delayed β -amylase activation, starch degradation, and sucrose formation (Purgatto *et al.*, 2001). Auxin-related signaling components also respond to ethylene during banana ripening, providing a biological basis for possible auxin–ethylene crosstalk (Kong *et al.*, 2022). Nevertheless, the present antioxidant analyses do not demonstrate that crude bacterial IAA delayed ethylene production, reduced ROS generation, or directly regulated phenylpropanoid metabolism. Therefore, crude bacterial auxin-mediated regulation of ripening and redox metabolism should be presented only as a mechanistic hypothesis.

Interpretation is further limited by the composition of the bacterial IAA solution. The nominal concentrations of 250, 500, and 1,000 ppm were determined using the Salkowski assay, whereas HPLC analysis of the corresponding ethyl acetate-extracted

Table 7 Total phenolic content (mg GAE/100 g FW) of ‘Kluai Khai’ banana treated with different concentrations of crude bacterial IAA under temperature phasing (TP) and non-TP conditions during 10 days of storage

Treatments		Total phenolic content (mg GAE/100 g FW)				
IAA conc. (ppm)	Temp. phasing	Day 0	Day 1	Day 6	Day 8	Day 10
0	Non-TP	28.66 $\pm$ 1.0	23.70 $\pm$ 1.1 ^b	26.76 $\pm$ 0.22 ^c	25.74 $\pm$ 0.33 ^f	31.64 $\pm$ 2.78
	TP	28.66 $\pm$ 1.0	21.80 $\pm$ 0.5 ^{cd}	33.35 $\pm$ 0.94 ^a	31.82 $\pm$ 0.28 ^b	38.36 $\pm$ 1.46
250	Non-TP	28.66 $\pm$ 1.0	22.64 $\pm$ 0.2 ^c	22.19 $\pm$ 0.54 ^e	28.21 $\pm$ 0.37 ^e	26.66 $\pm$ 0.32
	TP	28.66 $\pm$ 1.0	18.95 $\pm$ 0.4 ^e	24.88 $\pm$ 0.27 ^d	30.74 $\pm$ 0.24 ^c	32.50 $\pm$ 0.45
500	Non-TP	28.66 $\pm$ 1.0	24.43 $\pm$ 0.4 ^b	28.62 $\pm$ 0.28 ^b	28.73 $\pm$ 0.17 ^e	33.74 $\pm$ 0.29
	TP	28.66 $\pm$ 1.0	25.94 $\pm$ 0.4 ^a	25.84 $\pm$ 0.43 ^{cd}	35.56 $\pm$ 0.42 ^a	38.04 $\pm$ 0.54
1,000	Non-TP	28.66 $\pm$ 1.0	24.27 $\pm$ 0.2 ^b	25.55 $\pm$ 0.56 ^d	29.17 $\pm$ 0.29 ^d	35.29 $\pm$ 0.35
	TP	28.66 $\pm$ 1.0	21.34 $\pm$ 0.4 ^d	27.80 $\pm$ 0.76 ^b	35.76 $\pm$ 0.63 ^a	38.77 $\pm$ 0.68
Crude IAA		-	**	**	**	**
TP		-	**	**	**	**
IAA $\times$ TP		-	**	**	**	ns
CV (%)		-	2.24	2.17	1.04	3.05

Note: Means within the same column followed by different superscript letters (a–f) are significantly different according to the least significant difference (LSD) test at $P < 0.05$. Values are presented as mean $\pm$ standard deviation ($n = 3$). ** $P < 0.01$, ns = not significant. IAA conc. = IAA concentration, Temp. phasing = temperature phasing, CV = coefficient of variation.

fraction indicated that authentic IAA represented approximately 43.86% of the Salkowski-estimated concentration. Based on this comparison, the estimated authentic IAA concentrations were approximately 109.65 ± 3.24 , 219.29 ± 6.47 , and 438.58 ± 12.95 ppm, respectively. The Salkowski reagent can react with several indolic compounds and is therefore not specific to IAA (Glickmann and Dessaux, 1995). However, the inconsistent responses among concentrations cannot be attributed solely to the lower-than-expected levels of authentic IAA, because the crude solution contained other uncharacterized metabolites that may also have influenced fruit ripening. Future studies should therefore isolate and purify bacterially derived IAA, characterize other metabolites present in the crude solution, and measure ethylene production, ROS accumulation, and relevant enzyme activities.

CONCLUSIONS

Temperature phasing was the principal treatment associated with maintaining the postharvest quality of 'Kluai Khai' bananas. All fruit reached the fully yellow stage by day 6 and showed comparable total soluble solids at the end of storage, indicating that TP did not prevent normal ripening. Its benefits became most evident during the later stages, when TP-treated fruit retained brighter and more uniformly yellow peel, greater peel-inclusive firmness, higher titratable acidity, and lower weight loss than fruit maintained continuously at 25°C. The combination of TP and the 1,000 ppm crude bacterial solution maintained acceptable visual quality through day 10, approximately four days longer than the untreated non-TP control. However, TP modified the temporal pattern of respiration rather than consistently suppressing respiratory activity.

Antioxidant-related responses were assay- and stage-dependent. TP-treated fruit generally exhibited higher FRAP values and total phenolic content during

late storage, whereas DPPH radical-scavenging activity was higher in non-TP fruit. These contrasting patterns indicate that TP may have altered the composition or relative contribution of extractable antioxidants rather than uniformly enhancing overall antioxidant activity. The effects of the crude bacterial IAA solution varied among concentrations, temperature conditions, sampling dates, and measured attributes. Consequently, the results do not demonstrate a consistent additional benefit of crude bacterial IAA or provide general evidence of synergy between IAA and TP.

Interpretation of the bacterial treatment is further constrained by the non-specificity of the Salkowski assay, the finding that authentic IAA represented approximately 43.86% of the nominal IAA-equivalent concentration, and the possible presence of other uncharacterized metabolites. The observed responses therefore cannot be attributed exclusively to IAA. Overall, TP alone represents the most evidence-supported practical treatment for delaying late-stage quality deterioration in 'Kluai Khai' bananas. Future studies should use purified or chromatographically standardized bacterial IAA, characterize other metabolites in the crude preparation, and directly measure ethylene production, ROS accumulation, PPO activity, and antioxidant- and softening-related enzymes to clarify the physiological mechanisms involved.

CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest related to this research.

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